

DAC Enode Intelligence Watcher

Automated Observation, Rotation Intelligence & Report Pipeline — v1.9.3

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Data Source	enodes.dachain.tech/testnet · watcher-generated JSON snapshots · manual backfill archive · rotation intelligence summary · anomaly detection summary · concentration risk summary · dashboard / export outputs
Report Type	Infrastructure Automation Analysis — Enode Rotation Intelligence, Dashboarding & Report Pipeline
Relates to	Official Enode Evolution Analysis — Infrastructure Rotation & Network Maturation (11 May 2026)
Report Archive	https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/Testnet (Inception) Technical Reports

Executive Summary

This report documents the evolution of the DAC Enode Intelligence Watcher from a basic official enode monitoring script into a complete infrastructure observation, intelligence, dashboard, and report export pipeline. The previous report — "Official Enode Evolution Analysis — Infrastructure Rotation & Network Maturation" (May 11, 2026) — established that the DAC Testnet official enode list is a living operational configuration rather than a static endpoint. This report continues that work by documenting how the observation process was automated, enriched, and converted into a reusable reporting system.

Metric	Value
Total observations	22
Manual backfill observations	14
Automated watcher observations	8
First observed source time	May 15, 2026 (00:00 CEST)
Latest observed source time	Jun 3, 2026 (12:00 CEST)
Minimum observed enode count	7
Maximum observed enode count	15
Average observed enode count	12.0
Total added events	58
Total removed events	33
Target port observed	28657

Selected anomaly signals	5
Highest anomaly severity	HIGH
Current concentration label	ELEVATED
Current export formats	Markdown · JSON · PDF

Latest observation (Jun 3, 2026 — 12:00 CEST): current enodes = 10 · previous = 14 · removed = 4 · added = 0 · change severity = MEDIUM. This is an operational snapshot captured by the watcher, not a final network topology.

Observation Methodology

The watcher monitors the official DAC Testnet public enode page at <https://enodes.dachain.tech/testnet/>. Each scheduled run extracts the source timestamp, target port, node IDs, IP addresses, and ports — then compares the current state against the previously stored latest.json baseline. Changes are classified by added, removed, unchanged, change severity, and severity reason.

Source	Official public DAC Testnet enode page
Schedule	Every 15 minutes via GitHub Actions
Comparison baseline	data/latest.json
Snapshot policy	Create a snapshot when meaningful change is detected
Ordering fields	source generated time · checked_at_utc · snapshot timestamp
Interpretation model	Deterministic, rule-based infrastructure observation

The watcher interval was changed from every 3 hours to every 15 minutes because the official enode list was observed to update more than once per day. A shorter interval reduces the chance of missing short rotation windows.

From Manual Observation to Automated Evidence

The earlier manual process depended on screenshots and copied text from the public enode page. That approach preserved point-in-time evidence but could miss updates when the source page changed between checks. DAC Enode Intelligence Watcher converts this into a repeatable data pipeline. Manual backfill observations remain preserved as historical evidence, while automated watcher snapshots continue the observation process from June 2026 onward.

```
Manual observation
↓
Risk of missed updates
↓
Automated watcher (every 15 minutes)
↓
Structured JSON snapshots
↓
Intelligence summaries
↓
```

Architecture & Data Flow Topology — v1.9.3

The current v1.9.3 architecture is a full observation, enrichment, reporting, dashboard, and export pipeline. Every layer produces a structured artifact used downstream.



Implementation Evolution

The project was built incrementally. Each update added one layer of observability, analysis, reporting, or usability rather than replacing the previous system.

Version	Layer	Main Artifact	Report-Use Value
v1.1–v1.2	Change Severity + Summary	change_severity, severity_reason, ai_style_summary	Converts raw changes into report-ready interpretation
v1.3	Enode Rotation Intelligence	build_rotation_intelligence.py; rotation-intelligence-summary.json	Aggregates manual + automated observations into one timeline
v1.4	Anomaly Detection	build_anomaly_detection.py; anomaly-detection-summary.json	Flags unusual rotation events as candidates for review
v1.5	Technical Report Generator	generate_technical_report.py; dac-enode-intelligence-report.md	Creates reusable Markdown report drafts
v1.6	Static Dashboard Layer	dashboard/index.html	Makes watcher output visually inspectable

v1.6.1	Provider / ASN Hint Layer	provider_hints.py	Adds static provider / ASN context
v1.6.2	DAC Infrastructure Signal Layer	dac_signal_hints.py	Adds community inference labels with disclaimers
v1.6.3	Live ASN Lookup Option	asn_lookup.py; asn-cache.json	Adds live routing/provider enrichment with cache fallback
v1.7	Provider Concentration Risk	build_concentration_risk.py; concentration-risk-summary.json	Adds observation-based concentration context
v1.8.0	Core Dashboard Charts	Enode count and added/removed charts	Shows movement visually
v1.8.1	Distribution Charts	Live ASN and DAC Infrastructure Signal distribution	Shows concentration and signal spread visually
v1.8.2	Summary Charts	Anomaly severity and manual-vs-automated coverage	Completes first chart layer
v1.9.0	Custom Markdown Export	generate_custom_report.py	Adds 7D, 30D, ALL TIME Markdown reports
v1.9.1	Custom JSON Export	--format json / both	Adds structured outputs for AI and scripts
v1.9.2	Dashboard Export Links	Static links + browser custom range export	Makes dashboard a report access surface
v1.9.3	Optional PDF Export	generate_pdf_report.py; reportlab	Completes Markdown / JSON / PDF export stack

Observation Dataset Summary

The all-time dataset combines manual backfill observations and automated watcher observations, making it possible to compare the pre-watcher period against the live scheduled observation period while keeping the two phases clearly separated.

Phase	Observation Count	Description
Manual backfill	14	Historical pre-watcher observations converted into structured JSON
Automated watcher	8	Scheduled GitHub Actions watcher observations from June 2026 onward
Total	22	Combined all-time observation dataset

Enode Count Movement

Across the all-time dataset, the observed official enode count moved between 7 and 15. This confirms that the official enode registry remains a living configuration, while the target P2P port stayed consistent at 28657 throughout.

Minimum enode count	7
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Maximum enode count	15
Average enode count	12.0
Total added events	58
Total removed events	33
Target port observed	28657 (consistent across all 22 observations)

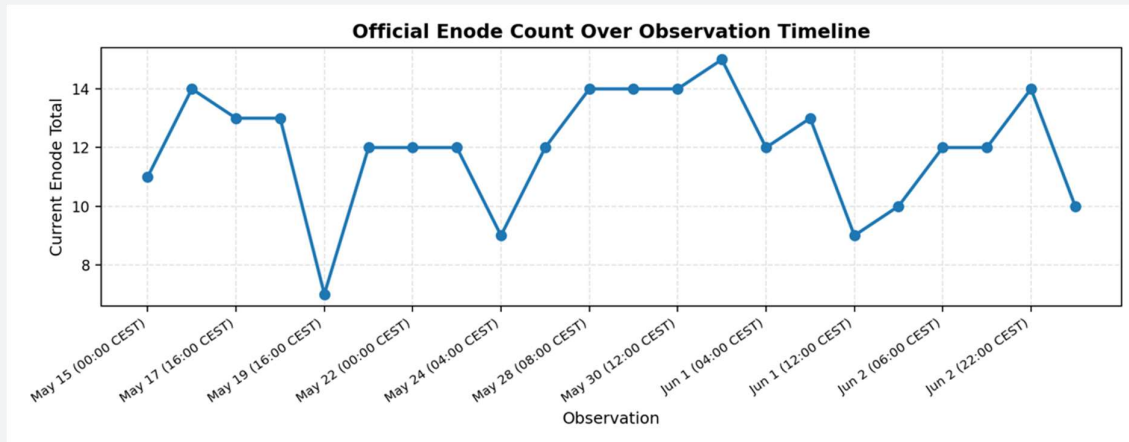


Fig. 1 — Official Enode Count Over Observation Timeline

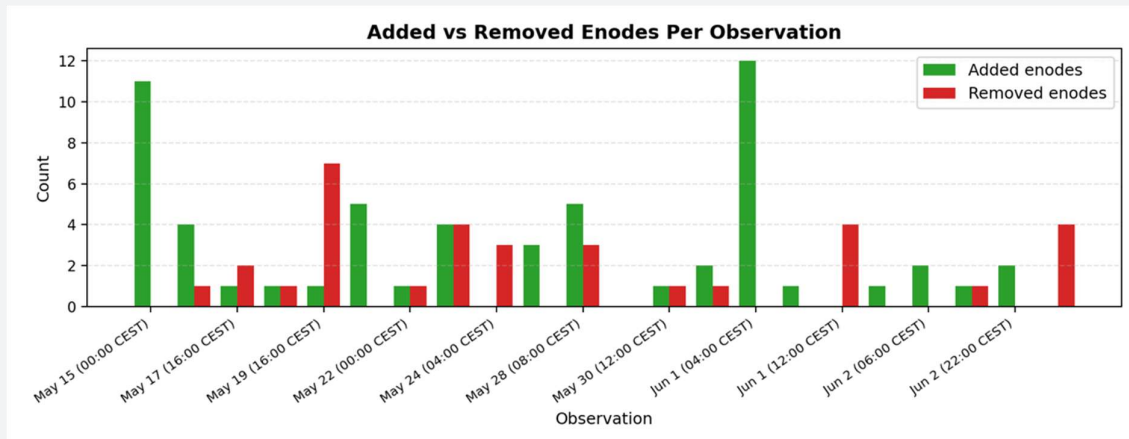


Fig. 2 — Added vs Removed Enodes Per Observation

Observation Timeline

The table below summarizes the all-time observation timeline preserved by the watcher and manual backfill archive. Manual backfill observations do not carry a checked_at_utc because they predate automation deployment.

#	Source Time	Phase	Status	Curr	+	-	=	Port
1	May 15, 2026 (00:00 CEST)	Manual	Initial	11	11	0	0	28657
2	May 16, 2026 (20:00 CEST)	Manual	Changed	14	4	1	10	28657
3	May 17, 2026 (16:00 CEST)	Manual	Changed	13	1	2	12	28657
4	May 18, 2026 (08:00 CEST)	Manual	Changed	13	1	1	12	28657
5	May 19, 2026 (16:00 CEST)	Manual	Changed 	7	1	7	6	28657
6	May 20, 2026 (20:00 CEST)	Manual	Changed	12	5	0	7	28657
7	May 22, 2026 (00:00 CEST)	Manual	Changed	12	1	1	11	28657
8	May 23, 2026 (12:00 CEST)	Manual	Changed 	12	4	4	8	28657
9	May 24, 2026 (04:00 CEST)	Manual	Changed	9	0	3	9	28657
10	May 25, 2026 (04:00 CEST)	Manual	Changed	12	3	0	9	28657
11	May 28, 2026 (08:00 CEST)	Manual	Changed 	14	5	3	9	28657
12	May 29, 2026 (04:00 CEST)	Manual	Unchanged	14	0	0	14	28657
13	May 30, 2026 (12:00 CEST)	Manual	Changed	14	1	1	13	28657
14	May 31, 2026 (12:00 CEST)	Manual	Changed	15	2	1	13	28657
15	Jun 1, 2026 (04:00 CEST)	Auto	Initial	12	12	0	0	28657
16	Jun 1, 2026 (08:00 CEST)	Auto	Changed	13	1	0	12	28657
17	Jun 1, 2026 (12:00 CEST)	Auto	Changed	9	0	4	9	28657
18	Jun 1, 2026 (22:00 CEST)	Auto	Changed	10	1	0	9	28657
19	Jun 2, 2026 (06:00 CEST)	Auto	Changed	12	2	0	10	28657
20	Jun 2, 2026 (12:00 CEST)	Auto	Changed	12	1	1	11	28657
21	Jun 2, 2026 (22:00 CEST)	Auto	Changed	14	2	0	12	28657
22	Jun 3, 2026 (12:00 CEST)	Auto	Changed	10	0	4	10	28657

 = anomaly signal detected at this observation.

Anomaly Detection Summary

The anomaly layer is deterministic and rule-based. It highlights observation candidates for deeper review. It does not automatically classify the network as unhealthy.

Selected anomaly signals	5
Highest severity	HIGH
Severity counts	HIGH = 5
Anomaly type counts	LARGE_ENODE_COUNT_DROP = 1 · HIGH_REMOVAL_EVENT = 1 · AGGRESSIVE_ROTATION = 3
Recommended action	Use these anomaly events as candidates for deeper manual review and future technical reporting.

Obs.	Source Time	Type	Severity	Reason
5	May 19, 2026 (16:00 CEST)	LARGE_ENODE_COUNT_DROP	HIGH	Official enode count decreased by 6 — from 13 to 7.
5	May 19, 2026 (16:00 CEST)	HIGH_REMOVAL_EVENT	HIGH	7 enodes were removed from a previous total of 13.
5	May 19, 2026 (16:00 CEST)	AGGRESSIVE_ROTATION	HIGH	Large rotation intensity: 1 added and 7 removed.
8	May 23, 2026 (12:00 CEST)	AGGRESSIVE_ROTATION	HIGH	Large rotation intensity: 4 added and 4 removed.
11	May 28, 2026 (08:00 CEST)	AGGRESSIVE_ROTATION	HIGH	Large rotation intensity: 5 added and 3 removed.

Anomaly signals indicate observation candidates for review. They do not independently prove network failure or official operational issues.

Provider / ASN Concentration and DAC Infrastructure Signal Context

The concentration layer provides an observation-based heuristic for understanding whether observed enode IPs appear concentrated across a small number of ASNs, countries, providers, or DAC Infrastructure Signal categories. This layer should be read as operational context, not as definitive decentralization measurement or official DAC ownership classification.

Overall concentration label	ELEVATED
Headline	Observed infrastructure shows elevated concentration under the current heuristic model.
Key observation	Top live ASN is AS51167 with 15 unique IPs (51.72%).
Country observation	Top live ASN country code is DE with 18 unique IPs (62.07%).

Interpretation	Top live ASN controls at least 50% of observed unique IPs. Observed IPs show notable concentration in one live ASN country code.
Recommended action	Use as an observation aid only. Compare with registry history, DAC Infrastructure Signal, manual peer identity evidence, live ASN lookup updates, and future watcher snapshots before drawing conclusions.

Dimension	Top Name	Top Count	Top %	Label
Live ASN	AS51167	15	51.72%	ELEVATED
Live Country	DE (Germany)	18	62.07%	ELEVATED
DAC Infrastructure Signal	Unknown / No Signal	9	31.03%	LOW

Provider concentration and decentralization risk summary is an observation-based heuristic. It is based on currently available watcher data, live ASN enrichment, static provider hints, and DAC Infrastructure Signal labels. It should not be treated as an official DAC classification or as a definitive decentralization measurement.

Dashboard and Report Export Pipeline

By v1.9.3, the watcher is no longer only a monitoring bot. It is a dashboard-backed report preparation system. The dashboard exposes all watcher intelligence layers through visual charts and structured export links.

Dashboard Layers

Layer	Current Function
Dashboard	Displays latest watcher state, summaries, charts, persistent entities, anomaly events, and observation timeline
Core charts	Enode Count Over Time and Added vs Removed Per Observation
Distribution charts	Live ASN Distribution and DAC Infrastructure Signal Distribution
Summary charts	Anomaly Severity Count and Manual vs Automated Observations
Static exports	7D / 30D / ALL TIME Markdown, JSON, and PDF links
Custom browser export	User-selected From / To observation window exported as Markdown, JSON, or browser PDF

Export Formats

Format	Purpose	Generated Output
Markdown	Editable human-readable report source	dac-enode-report-7d.md · 30d.md · all.md
JSON	Structured AI-assisted analysis and scripts	dac-enode-report-7d.json · 30d.json · all.json
PDF	Optional final packaging for sharing or presentation	dac-enode-report-7d.pdf · 30d.pdf · all.pdf

Key Findings

The official enode list remains a living configuration.

The all-time observation timeline shows repeated enode count changes and rotation events across 22 observations, extending from 7 to 15 enodes. This extends the earlier finding from Report 5 that the official registry should be treated as an operational reference, not as a permanent peer list.

Target port consistency is a strong operational signal.

Across the entire all-time dataset, the target port remained 28657 without exception. While peer membership rotated frequently, the public P2P access convention remained stable.

Automation improves evidence quality.

The 15-minute watcher schedule, latest-state tracking, and snapshot preservation reduce dependence on manual screenshots and help preserve short-lived rotation windows. Manual backfill observations confirmed 14 distinct state points that would have required manual effort to preserve.

Anomaly detection should be used as a review layer, not a verdict.

The current anomaly layer found five HIGH signals, primarily around Observation 5 (May 19 — count drop from 13 to 7). These are useful candidates for deeper review but do not independently indicate network failure or degraded operation.

Provider concentration context requires careful interpretation.

The current heuristic model labels concentration as ELEVATED, with AS51167 and DE appearing as top dimensions across the all-time dataset. This is useful operational context but not definitive ownership or decentralization proof.

The watcher has become a reusable reporting system.

The project now generates Markdown, JSON, and PDF outputs and exposes them through dashboard links and custom browser exports, creating a self-contained infrastructure intelligence tool.

Implications for Node Operators

The watcher reinforces the operational lesson from the earlier enode evolution report: node operators should treat the official enode registry as a live reference, not a one-time configuration artifact.

- Monitor `enodes.dachain.tech/testnet` periodically, or deploy an automated watcher. The 15-minute interval confirmed that changes can occur within hours.
- Update static peer configuration (`static-nodes.json` and `config.toml`) when the official list changes significantly.
- Do not assume removed public enodes are permanently offline — removal from the published list may only reflect bootstrap-list curation, not node shutdown.
- Keep local operational evidence separate from official source evidence — local RPC and explorer data remain the primary source of truth for individual node status.
- Use anomaly signals as review prompts, not automatic conclusions. Aggressive rotation events may reflect routine infrastructure management.
- Use Markdown and JSON exports as primary analysis material; use PDF as final packaging only when needed.

Limitations and Interpretation Boundaries

- The watcher observes only publicly available enode data from the official source and locally generated summaries.
- Provider and ASN labels are enrichment hints based on routing or heuristic data — they are not official DAC classifications.
- DAC Infrastructure Signal labels are community inference labels, not official DAC ownership classifications.
- Provider concentration labels are observation-based heuristics, not definitive decentralization measurements.
- Manual backfill data is historical and may contain gaps because it predates continuous automation.
- GitHub Actions scheduled runs may have slight delay — source timestamps and `checked_at_utc` should be used for ordering, not the watcher run time alone.

Future Outlook

The v1.9.3 export roadmap is complete, but the watcher can evolve into a broader infrastructure intelligence system if additional observation needs emerge.

Possible Upgrade	Use Case	Status
Public RPC health monitoring	Track availability and latency of public RPC endpoints	<i>Optional future direction</i>
Explorer availability monitoring	Detect explorer outages or slowdowns	<i>Optional future direction</i>
Multi-source infrastructure watcher	Correlate enode, RPC, explorer, and node telemetry signals	<i>Possible later</i>
Richer dashboard filtering	Filter by phase, severity, ASN, signal, or time window	<i>Possible later</i>
Automated comparison reports	Compare any two observation windows automatically	<i>Possible later</i>

Conclusion

The DAC Enode Intelligence Watcher represents the transition from manual infrastructure observation to continuous, structured, report-ready intelligence collection. The previous enode evolution report established that the DAC official enode registry is dynamic. This report demonstrates that the observation process has now matured into an automated system capable of preserving evidence, detecting rotation, enriching infrastructure context, visualizing trends, and producing reusable reports.

The current dataset shows that the official enode list continues to rotate while maintaining target-port consistency at 28657. Anomaly signals and concentration labels provide useful review cues, but they should remain bounded by the project disclaimer: they are observation-based heuristics, not official DAC classifications or definitive decentralization claims.

The key operational conclusion remains unchanged: the official enode registry should be treated as a living reference, and its evolution is part of the ongoing DAC Testnet infrastructure lifecycle.

Live Peer Topology — June 3, 2026 (10:57 UTC)

Active peer connectivity data was captured from both running nodes on June 3, 2026 at 10:57 UTC — at the time this report was being finalized. Both nodes were fully synced and actively connected to the DAC Testnet network. This data supplements the watcher dataset with real-time ground-truth peer evidence.

Windows node status	eth.syncing = false · net.peerCount = 17
WSL node status	eth.syncing = false · net.peerCount = 16
Static peer list	31 enodes (Windows + WSL internal peers + all accumulated external peers)

Active Peer Identity Map

The admin.peers output revealed the self-declared identities of each connected node. This observation extends the identity map from Report 5 (May 11) with several newly identified nodes and naming updates.

Node Identity	IP Address	Platform	Classification
DAC Testnet Authority 1	157.173.127.31	linux-amd64	Authority node (legacy build)
DAC Testnet RPC 02	157.173.127.22	linux-amd64	Official RPC node
DAC Testnet RPC 04	157.173.127.18	linux-amd64	Official RPC node
DAC-Node 05	206.189.127.204	linux-amd64	Official relay node
SAPInode	213.136.82.243	linux-amd64	Unlisted — active peer
whale-vps1	217.76.53.98	linux-amd64	Community VPS
whale-vps3	161.97.89.27	linux-amd64	Community VPS (removed from registry)
x0rabbitt	5.104.86.129	linux-amd64	Community node
rkbatman	173.249.42.5	linux-amd64	Community node
skarl_node_01	207.154.217.91	linux-amd64	Community node
NODERS	5.9.116.21	linux-amd64	Community node
TeoViTeoVi	113.173.209.213	linux-amd64	Community node
GauGauDAC	118.71.126.107	darwin-amd64	Community node — macOS
Fertal	95.216.70.180	linux-amd64	Community VPS (removed from registry)
(no identity)	173.212.217.213	linux-amd64	Anonymous node
(no identity)	194.60.201.112	linux-amd64	Anonymous node
JERUZSALEM WINDOWS	192.168.100.7	windows-amd64	Tester node — internal

JERUZSALEM LINUX	192.168.100.7	linux-amd64	Tester node — internal
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Key Discoveries from June 3 Peer Data

RPC 02 and RPC 04 identities confirmed. 157.173.127.22 is now identified as DAC Testnet RPC 02 and 157.173.127.18 as DAC Testnet RPC 04. Combined with RPC 03 (84.46.253.182, never published) documented in Report 5, this reveals the existence of a broader official RPC infrastructure layer beyond what is visible in the public enode registry.
GauGauDAC runs on darwin-amd64 (macOS). The node at 118.71.126.107 is the only darwin-amd64 (macOS) node observed in the network. All other identified nodes run linux-amd64 or windows-amd64. This is a notable exception in the peer topology.
Community operator diversity is growing. New community node names not seen in May: GauGauDAC, rkbatman, skarl_node_01, TeoViTeoVi, NODERS. This indicates increasing community participation in the DAC Testnet P2P layer.
Authority 1 (157.173.127.31) still active despite removal from registry. DAC Testnet Authority 1 continues peering with both tester nodes, confirming the earlier observation that removal from the public bootstrap list does not mean a node is shut down.
Static peer list has grown to 31 entries. The cumulative peer list maintained across accumulated observations now contains 31 enodes. This reflects the historical depth of the observation program and the breadth of network participation.

Community Resources

The watcher and related infrastructure tooling are part of the same open-source DAC Infra Tester contribution set.

CGNAT Dual Node Setup Repository: <https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup>

DAC Enode Intelligence Watcher Dashboard: <https://edlweiss186.github.io/dac-dual-node-cgnat-setup/Observation-for-Technical-Report/dac-enode-intelligence-watcher/dashboard/>

Official DAC Enode Source: <https://enodes.dachain.tech/testnet/>

Technical Reports Archive: [https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/Testnet \(Inception\) Technical Reports](https://github.com/EdLWEISS186/dac-dual-node-cgnat-setup/tree/main/Testnet%20(Inception)%20Technical%20Reports)